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Ritwik Jain

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Education

Delhi Technological University (Formerly DCE) — Delhi, India
*Bachelor of Technology - Information Technology – GPA: **9.40*** — *August. 2025 - Jun. 2029*

Shaheed Rajpal DAV Public School — Delhi, India
*CBSE Class XII - **96.4%*** — *Apr. 2023 - May. 2025*

Fr. Agnel School — Noida, India
*CBSE Class X – **94.6%*** — *Apr. 2022 - Mar. 2023*

Work Experience

Deloitte Touche Tohmatsu India LLP — Noida, India
AI/ML Intern — *April 2026 - July 2026*

- Engineered multiclass classification ML pipelines on AWS SageMaker, Glue, S3, and Athena
- Migrated **10+** notebook-based ML workflows to production-ready AWS Glue jobs, automating the entire process
- Maintained inference logic, configuration files, and data processing components supporting cloud-based ML models in production.

Achievements

Smart India Hackathon
Team Member

- Engineered a gamified sustainability platform that used image-based AI analysis for plant health detection and waste-reuse recommendations, reaching **20+** active users during the trial period
- Shortlisted among the **Top 100** teams in the college's SIH internal round.

GreenTag (APOGEE'26)
Team Leader

- Led a **5**-member team to build an AI-powered civic issue reporting platform enabling citizens to submit complaints via voice or text.
- Integrated a Telegram bot for complaint registration and status updates, onboarding **15+** users during the demo phase.
- Won 2nd Prize out of **20+** competing teams.

Fin'O Hack (DTU)
Participant

- Developed DrishtiPay, a Flutter-based UPI payment app for visually impaired users, tested with **10** users.
- Implemented voice-guided payments (Speech-to-Text/Text-to-Speech) with AI-assisted QR scanning and biometric authentication.

Projects

Digital Twin | *Python, Gemini, RAG, SQL, Pocket-TTS* | *GitHub* | *Demo*

- Designed a Digital Twin AI agent emulating a scientist's knowledge, reasoning style, and communication patterns using a memory-augmented, time-aware RAG pipeline with cross-encoder reranking.
- Implemented zero-shot voice cloning (Pocket TTS) to generate speech from a 30-second reference sample, achieving **5**-second response latency.

Analyzing Autoencoders | *Pytorch, Deep Learning, Computer Vision* | *Github*

- Trained ANN- and CNN-based autoencoders from scratch in PyTorch across **8** architectural configurations, varying depth, bottleneck size, and skip connections.
- Achieved a best validation MSE of 0.000023 with a deep CNN autoencoder, a **99.86**% improvement over the baseline ANN autoencoder.

Technical Skills

Languages: Python, C, C++, HTML, CSS, SQLite
Frameworks/Libraries: MERN Stack, OpenCV, MediaPipe, TensorFlow/Keras, PyTorch, Autoencoders, Transformers
Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Arduino, Linux, Google Colab, AWS Athena, SageMaker, S3

Certifications

Deep Learning Specialization (DeepLearning.AI, Stanford University) | *View Certificate* — *Sep. 2025*
Machine Learning Specialization (DeepLearning.AI, Stanford University) | *View Certificate* — *Jan. 2026*
Google AI Intensive Course | *View Certificate* — *Oct. 2025*
British Airways Data Science Simulation | *View Certificate* — *Oct. 2025*
McKinsey Forward Program | *View Certificate* — *Oct. 2025*

Extra-Curricular Activities

Member, AIMS-DTU — Sep 2025 – Present
Contribute to AI/ML research projects and paper discussions in deep learning and generative AI.

Member, Sahitya DebSoc (Debating Society) — Oct 2025 – Present